Oluwaseyifunmi Olowe

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Education

University of Southern California — Los Angeles, CA
Bachelor of Science in Computer Engineering and Computer Science — *Expected May 2029*

- **Relevant Coursework:** Embedded Systems, Distributed Systems/IoT, Analog Circuits, Data Structures and Algorithms, Object-Oriented Programming

Technical Skills

Languages: C, C++, Python, MATLAB, Assembly
Systems & Tools: Embedded Linux, FreeRTOS, Git, CUDA, PyTorch, CMake, Docker, AWS, KiCad, SolidWorks
Protocols & Peripherals: CAN, SPI, I2C, UART/USART, ADC, PWM, GPIO, TCP/IP, HTTP, MQTT
Lab Equipment: Oscilloscope, Power Supply, Signal Generator, Digital Multimeter, Spectrum Analyzer

Experience

Semiconductor Intern — Los Angeles, CA
California DREAMS — *May 2026 – Aug 2026*

- Designed a **4-layer** audio amplifier PCB in KiCad within a 2 in × 2 in footprint, from schematic to fabrication
- Designed and tested stereo-summing, **low-pass filtering**, gain, and **power-decoupling** circuits to amplify bass
- Verified frequency responses, gain, and supply voltages using an **oscilloscope, multimeter**, and signal generator
- Reworked a heart-rate monitor PCB using **SMD soldering** and identified **4 design errors** via schematic review

Embedded Systems Engineer — Los Angeles, CA
USC Formula SAE Electric — *Sep 2025 – Present*

- Develop STM32F4 accumulator-management firmware in **C/FreeRTOS** using CAN, SPI, **ADC-DMA**, flash, and watchdogs for real-time battery safety monitoring
- Build a **Python test-automation** GUI to decode DBC signals, replay CAN logs, and detect invalid data
- Use **MATLAB** to analyze cell voltage, temperature, and current telemetry and validate battery fault thresholds
- Write **7 host-side C test suites** with **100+ assertions** validating CAN protocols, battery logic, and ADC data

Undergraduate Researcher — Los Angeles, CA
University of Southern California — *May 2026 – Jul 2026*

- Optimized hardware-aware **spiking neural networks (SNNs)** for deployment on superconducting SFQ chips
- Pruned and quantized the network to fit a **3.4 × 3.9 mm** SFQ chip, **reducing fan-in and circuit complexity**
- Cut inference cost by **5×** (1.27M to 254K operations) with a **3.65% accuracy gain**, reducing on-chip energy use
- Built a reproducible PyTorch pipeline to evaluate SNN performance across **100+ seed-controlled experiments**

Projects

GPU ThermaGuard | *C++, ESP-IDF, FreeRTOS, CUDA, NVML, AWS, IoT*

- Built a **hardware-in-the-loop GPU validation platform** coupling an ESP32/FreeRTOS controller with an **NVIDIA T4** on AWS EC2
- Profiled GPU temperature, power, utilization, and clock behavior under **CUDA workloads** at 10 Hz, reducing peak temperature by **9°C**
- Implemented FreeRTOS networking tasks across **Wi-Fi, IPv4, TCP/TLS, and MQTT**
- Validated watchdogs and failsafe control through **50 injected thermal and communication faults**

Hawkeye | *Embedded C, FreeRTOS, Git, Linux*

- Built an autonomous multi-sensor fusion system in **Embedded C** to find significant light sources
- Fused phototransistor, MPU-6050 IMU, and temperature sensor data to drive a servo toward the target object
- Wrote a register-level **I2C driver** to interface the MPU-6050
- Architected a **FreeRTOS** task pipeline to run sensor data acquisition and servo motor control concurrently

Leadership & Involvement

Membership Chair — Los Angeles, CA
National Society of Black Engineers — *Jan 2026 – Present*

- Lead member recruitment and engagement initiatives to grow and retain NSBE's Black engineering community